ECE 329
Lecture 9

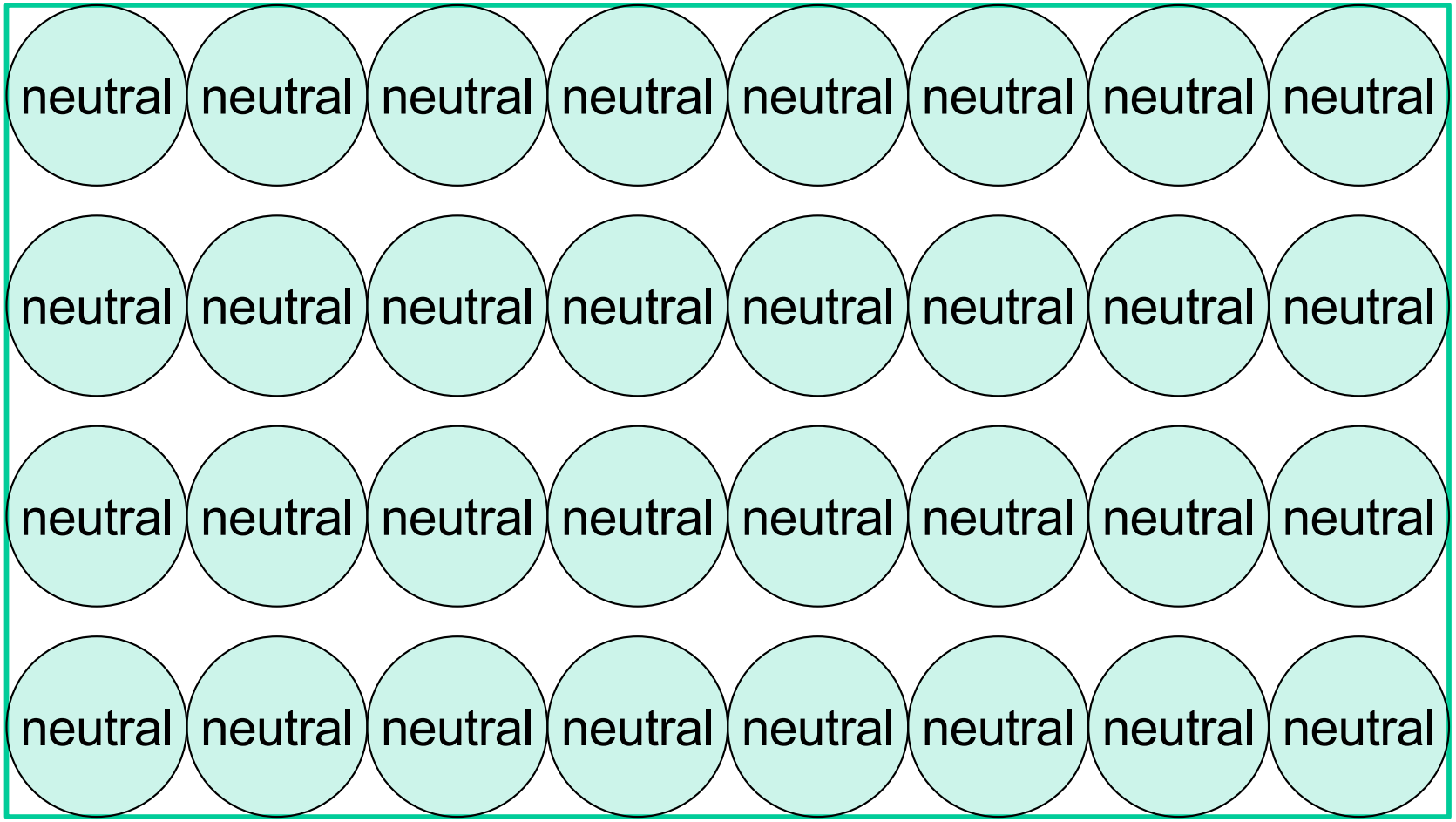
Dielectrics

Dr. Victoria Shao
yangshao@Illinois.edu

© Lynford Goddard, Brian Cunningham, Victoria Shao
- Copying, publishing or distributing without express written permission is prohibited.

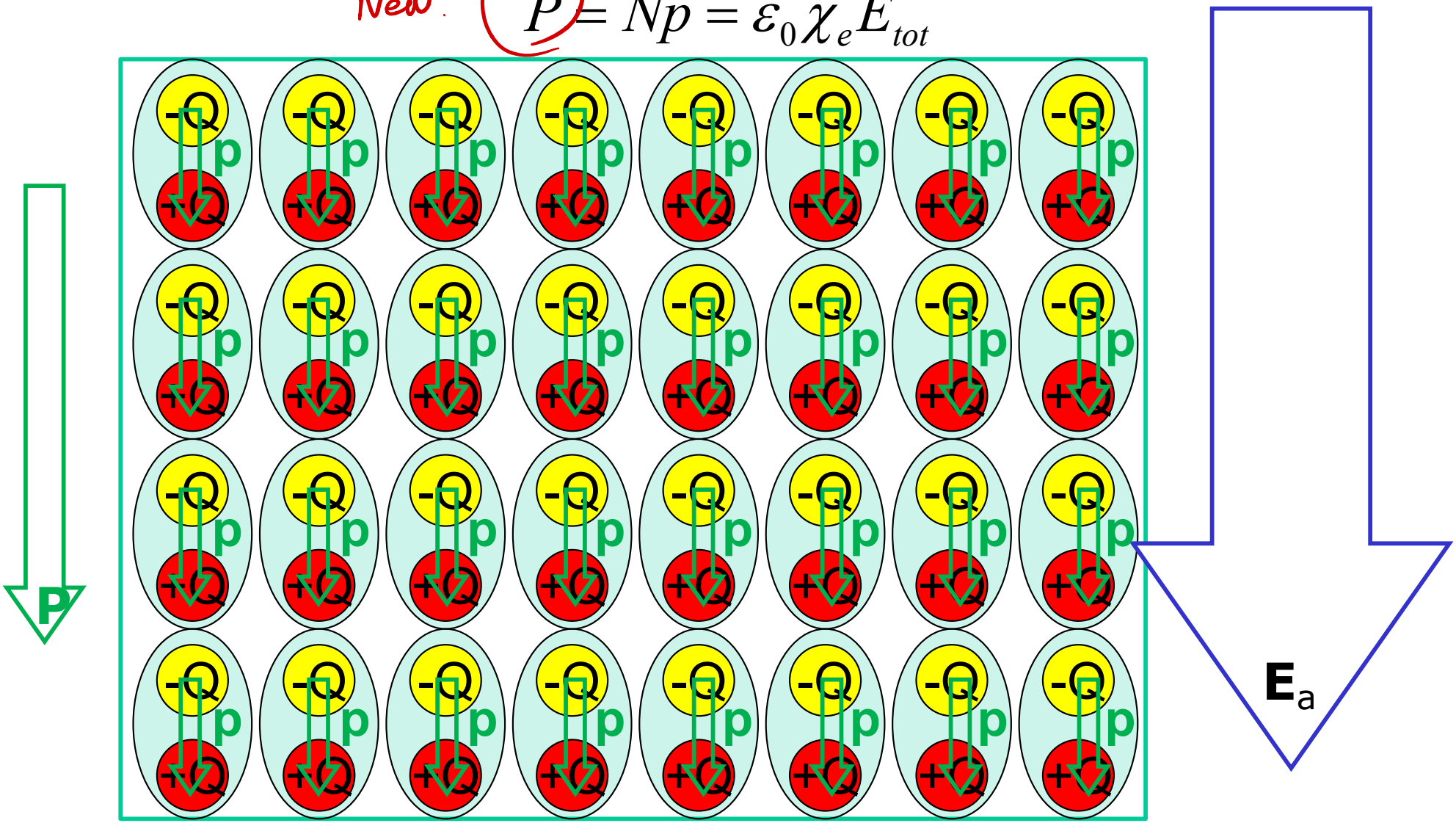
Apply **E** to Dielectric Material

Cancellation of internal charges



Apply **E** to Dielectric Material

New. $\vec{P} = N\vec{p} = \varepsilon_0 \chi_e \vec{E}_{tot}$



Dielectric Susceptibility χ_e

- Measures how easy it is to shift electrons from their centered orbit around the nuclei of a material to form internal dipoles

$$\vec{P} = \varepsilon_0 \chi_e \vec{E}_{tot}$$

$[C/m^2]$

unitless.

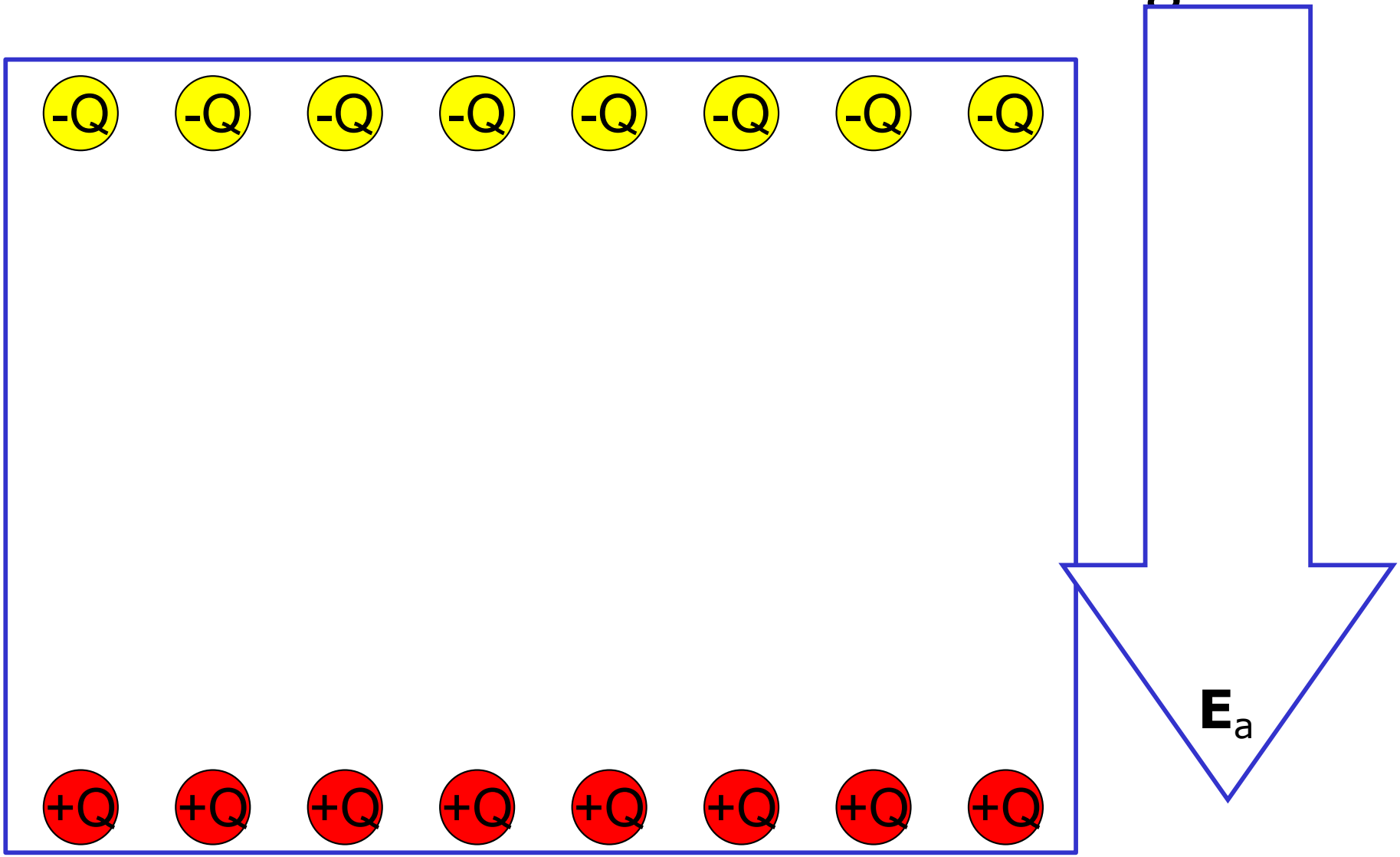
air $\vec{D} = \varepsilon_0 \vec{E}$

$[C/m^2]$ $[V/m]$

$[F/m]$

Apply **E** to Dielectric Material

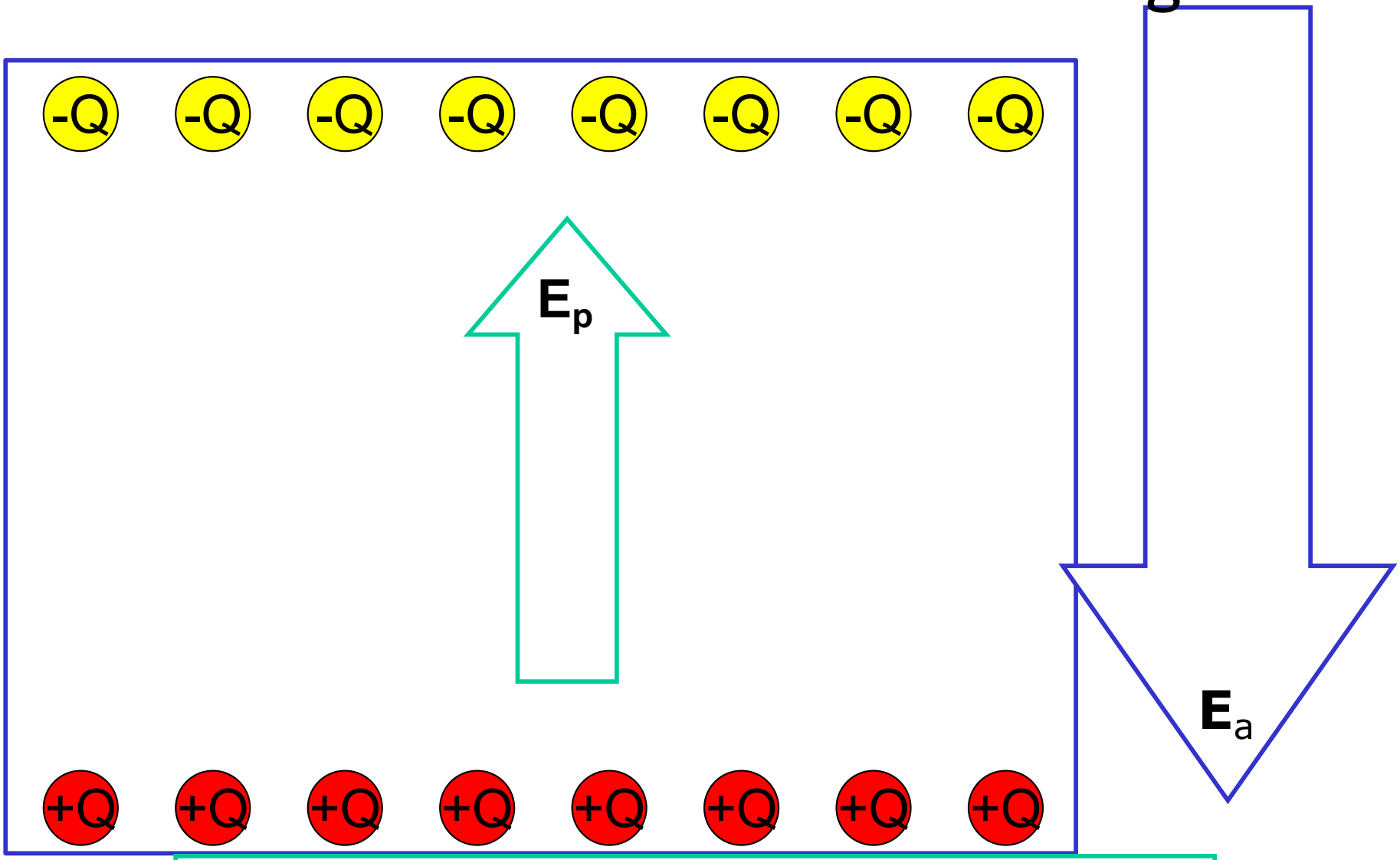
Cancellation of internal charges



"Polarization Surface Charges" Remain

Apply **E** to Dielectric Material

Cancellation of internal charges



Secondary Electric Field Produced

* *Does not cancel applied field inside dielectric*

$\vec{E}_{tot}$ is reduced but not zero

Unlike conductors where **E**=0, in the dielectric slab, the total field is **reduced (but NOT eliminated)** by the secondary field produced by the surface polarization charge

$$\vec{E}_{tot} = \vec{E}_a + \vec{E}_p$$

And the polarization vector, $\vec{P}$, is the polarization of atoms in the dielectric due to the TOTAL $\vec{E}_a$ field

$\vec{E}_a \rightarrow E_{tot}$

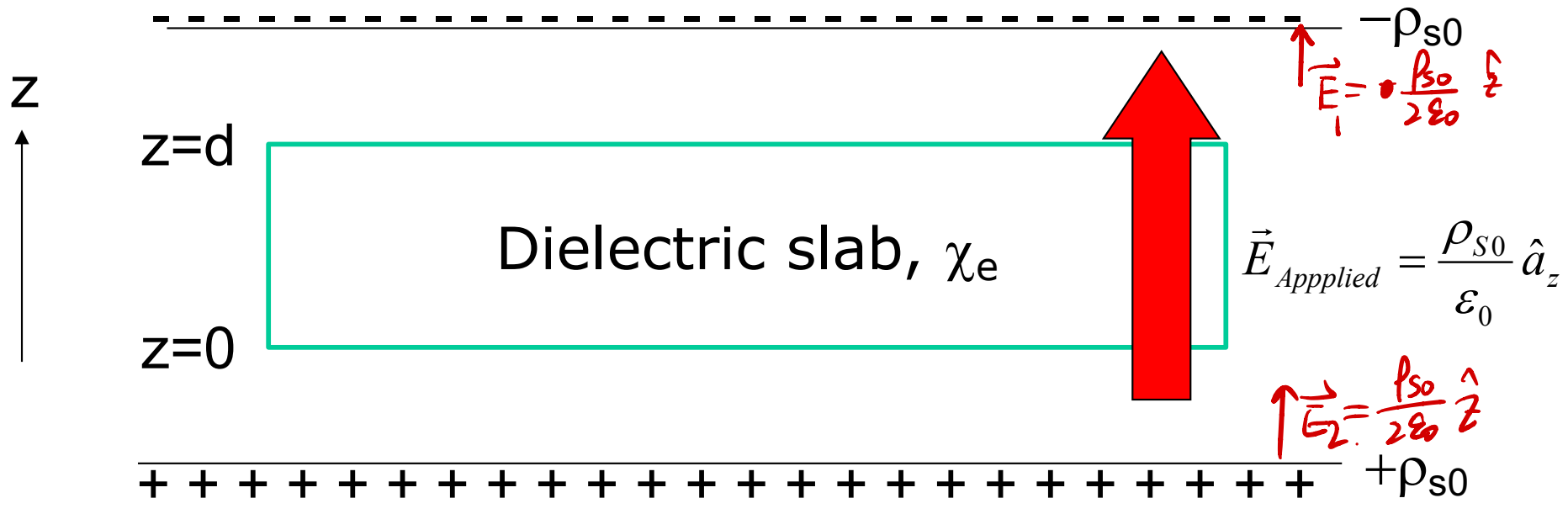
$$\vec{P} = \epsilon_0 \chi_e \vec{E}_{tot}$$

Electric susceptibility χ_e is a proportional constant.

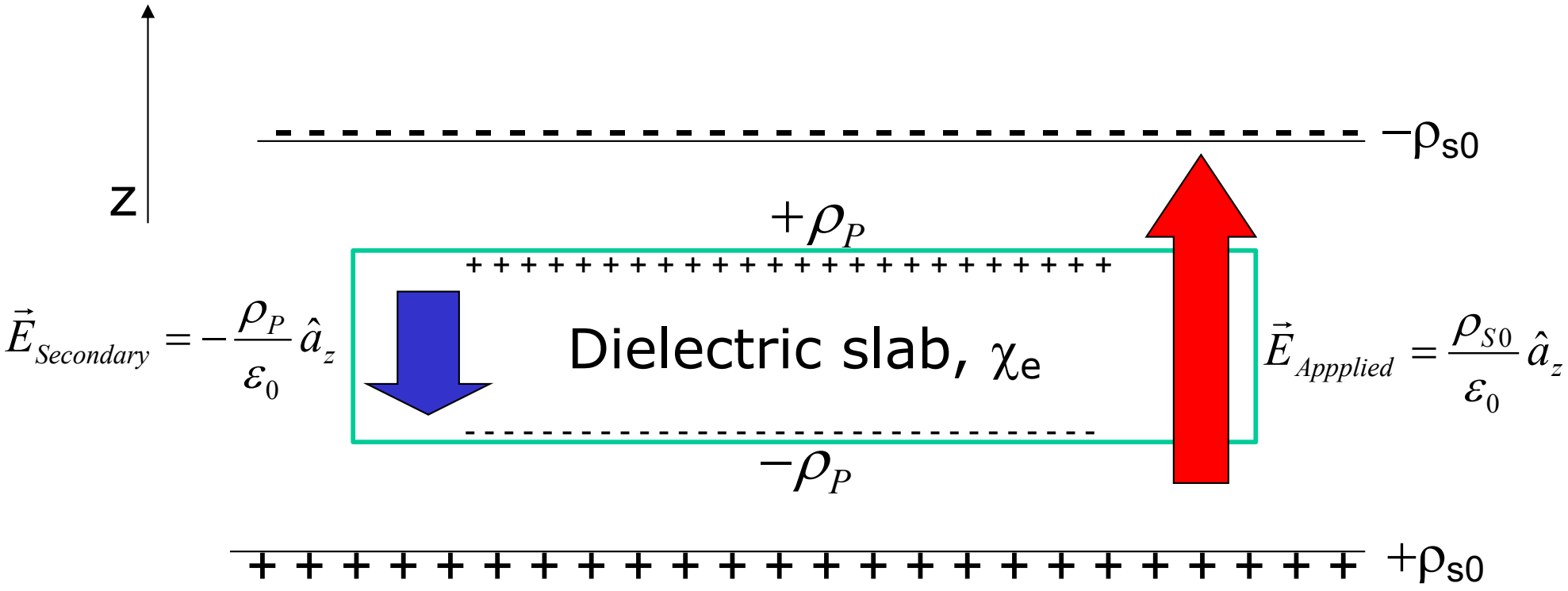
$\vec{P}$ represents the **dipole moment per unit volume** due to bound charge in the dielectric.

Example

- Apply external electric field to a slab of dielectric material
- Apply field by placing the slab between two equal and opposite charge densities



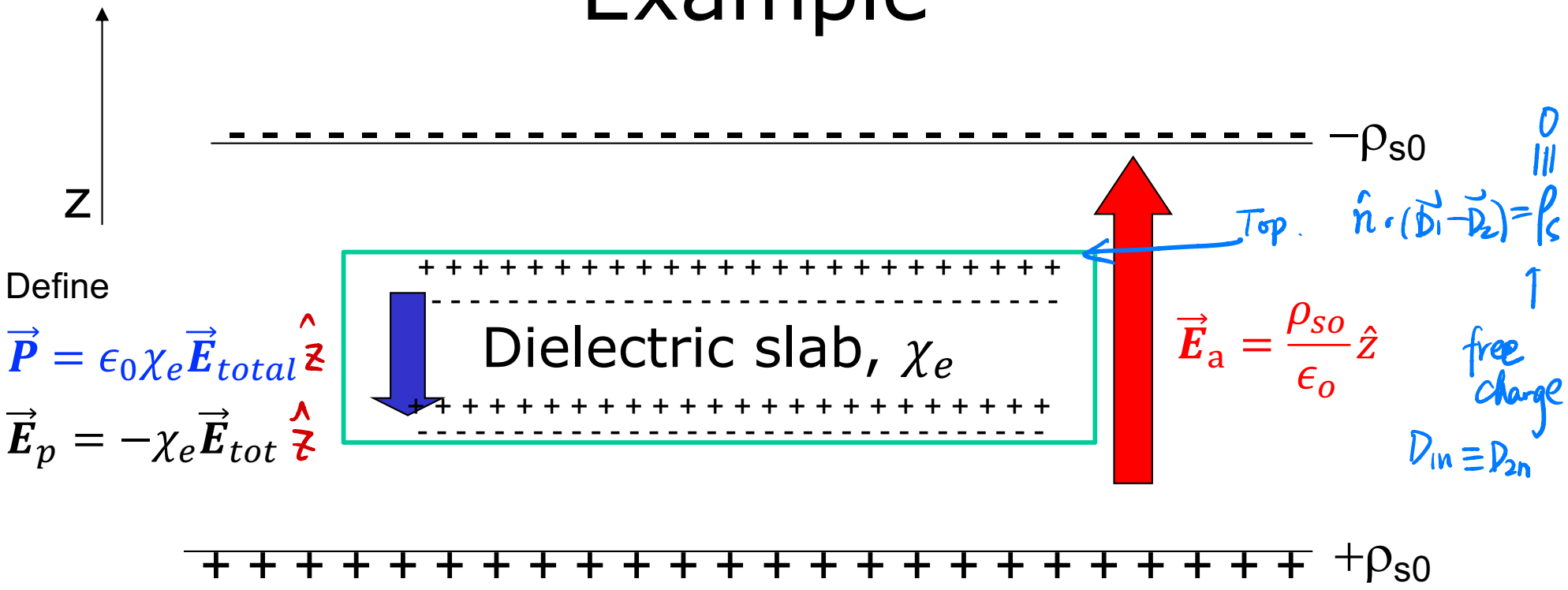
Example



$$\vec{E}_{total} = \vec{E}_{applied} + \vec{E}_{secondary}$$

The electric field inside a dielectric is **weaker than the applied field** because of polarization effects.

Example



$$\vec{E}_{tot} = \vec{E}_a + \vec{E}_p = \vec{E}_a - \chi_e \vec{E}_{tot}$$

$$\vec{E}_{tot} = \frac{\vec{E}_a}{1 + \chi_e}$$

$$\vec{D} \equiv \epsilon \vec{E}_{total} = \epsilon_0 (1 + \chi_e) \vec{E}_{total}$$

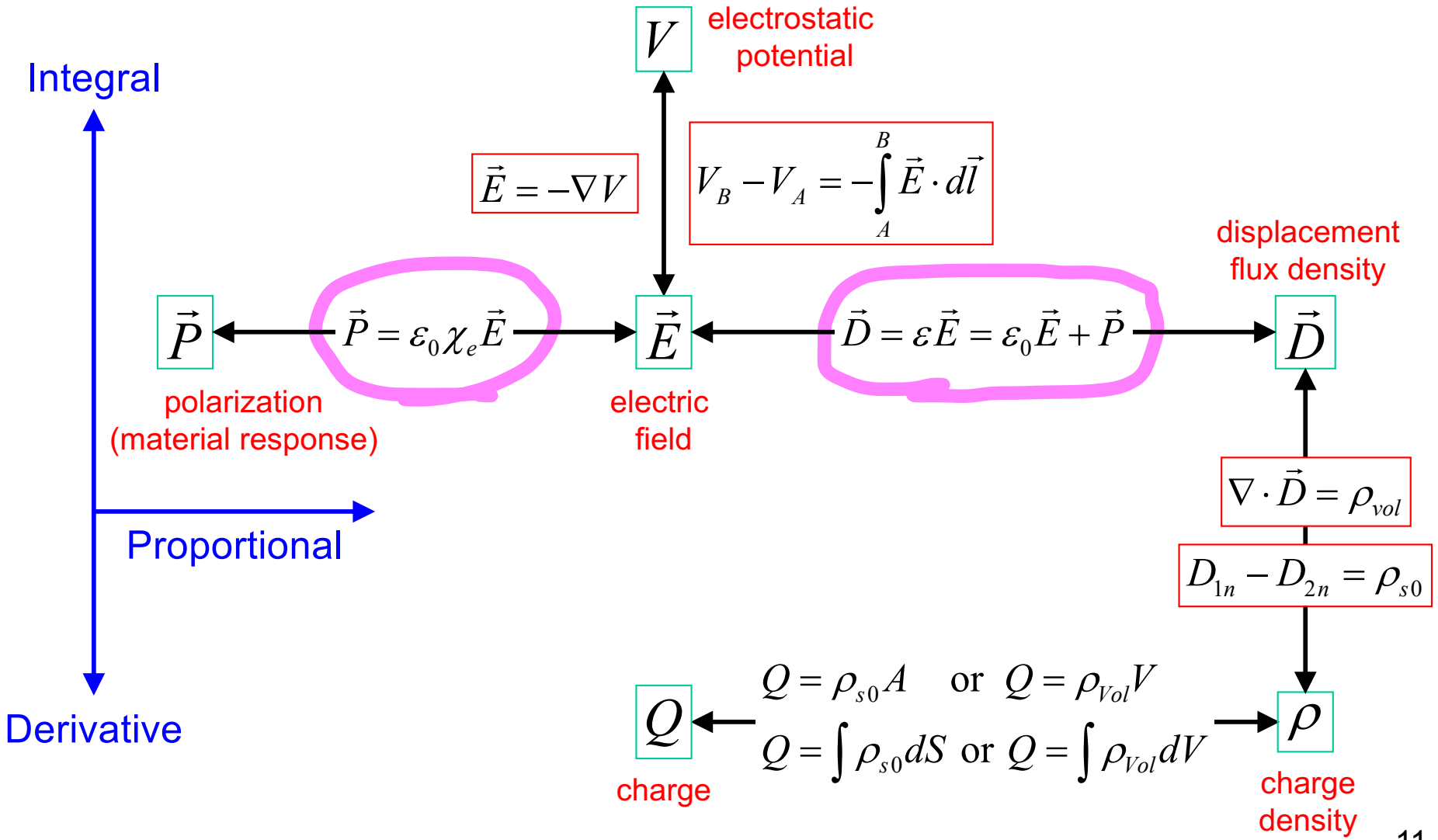
$$\vec{D} \equiv \epsilon_0 \vec{E}_{total} + \vec{P} = \rho_{so} \hat{z}$$

$\epsilon = \epsilon_0 \epsilon_r$

E-field strength reduced by $(1 + \chi_e)$

D-field strength is same as free space because charge is fixed

Connection of Concepts for Electrostatics



Example

So finally, the "final answer" is:

$$\vec{\boldsymbol{E}}_{tot} = \frac{\vec{\boldsymbol{E}}_a}{1+\chi_e} = \frac{\rho_{so}/\epsilon_o}{1+\chi_e} = \frac{\vec{\boldsymbol{D}}}{\epsilon}$$

$\vec{D} = \varepsilon \vec{E}$

The total electric field strength inside the dielectric is reduced from its "free space" value by $(1+\chi_e)$

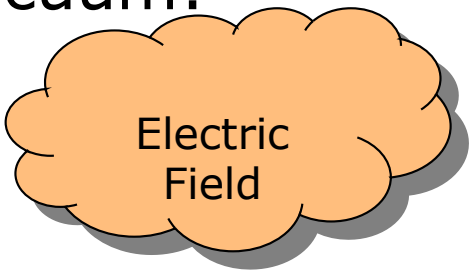
Free space: $\chi_e = 0$ Perfect Conductor: $\chi_e \rightarrow \infty$

no boundry charge

$\vec{P} = 0 \quad \vec{D} = \varepsilon_o \vec{E}$

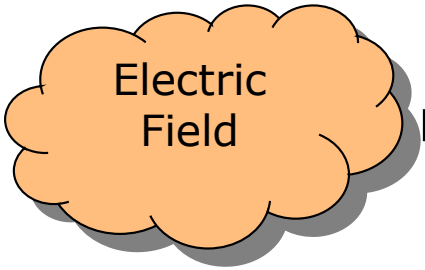
Displacement field $\vec{D}$

In vacuum:



Displacement
(electric flux density)
$\vec{D} = \epsilon_o \vec{\mathbf{E}}$

In dielectric:



Displacement
(electric flux density)
$\vec{D} = \epsilon_o \vec{\mathbf{E}} + \vec{P}$

Item $\epsilon_o \vec{\mathbf{E}}$: is the same as the case in vacuum

Item "$\vec{P} = \epsilon_0 \chi_e \vec{E}_{tot}$": is the polarization due to contribution of dipole formation in dielectric under applied $\vec{\mathbf{E}}$ field

Dielectric Constant ϵ

$$\vec{D} = \vec{P} + \epsilon_0 \vec{E} \qquad \vec{P} = \epsilon_0 \chi_e \vec{E}$$

$$\vec{D} = \epsilon_0 \chi_e \vec{E} + \epsilon_0 \vec{E}$$

$$\vec{D} = \epsilon_0 (1 + \chi_e) \vec{E}$$

$$\boxed{\vec{D} = \epsilon_0 \epsilon_r \vec{E}}$$

$\vec{E}$: total E field.

ϵ_r: Relative Permittivity or "Dielectric Constant" (unitless)

Dielectric Permittivity of the Material ϵ [F/m]

$$\boxed{\vec{D} = \epsilon \vec{E}}$$

$$\boxed{\varepsilon_r = (1 + \chi_e)}$$

$$\boxed{\varepsilon = \varepsilon_0 \varepsilon_r}$$

Units for ε_r: None

Units for χ_e: None

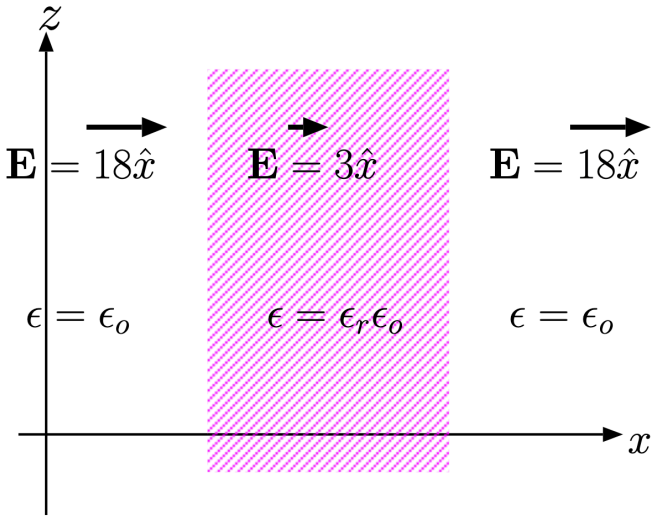
$\vec{D}, \vec{E}, \vec{P}$

Example: Infinite plane dielectric slab

An infinite plane dielectric slab lies in free space and has a constant electric field $\vec{E} = 18\hat{x}$ V/m in its exterior. Induced polarization of bound charges inside dielectric reduces the electric field strength inside the slab from $18\hat{x}$ V/m to $\vec{E} = 3\hat{x}$ V/m.

a) Find **D,** and **P** outside and inside the slab.

b) Calculate ϵ_r and χ_e.



a) $\vec{D} = \varepsilon\vec{E} = \begin{cases} \varepsilon_0\vec{E} = 18\varepsilon_0\hat{x} & \text{outside} \\ \varepsilon\vec{E} = 18\varepsilon_0\hat{x} & \text{inside} \end{cases}$ $[C/m^2]$

$\hat{n}\cdot(\vec{D}_1 - \vec{D}_2) = \rho_s = 0$ $\quad \vec{D}_{outside} = \vec{D}_{inside}$

$\vec{D} = \varepsilon_0\vec{E} + \vec{P}$

$\vec{P} = \vec{D} - \varepsilon_0\vec{E} = \begin{cases} 18\varepsilon_0\hat{x} - \varepsilon_0\cdot 18\hat{x} = 0 & \text{outside} \\ 18\varepsilon_0\hat{x} - \varepsilon_0\cdot 3\hat{x} = 15\varepsilon_0\hat{x} & \text{inside} \end{cases}$ $[C/m^2]$

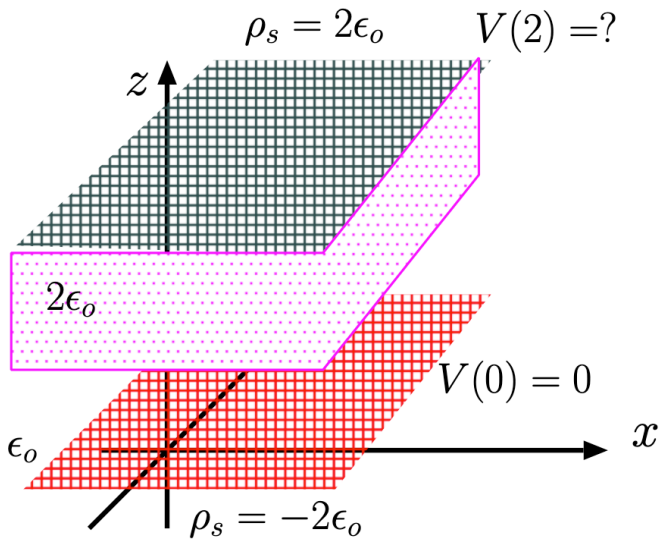
$\vec{P} = \varepsilon_0\chi_e\vec{E}$

b) $\varepsilon_0\varepsilon_r\cdot 3\hat{x} = 18\varepsilon_0\hat{x}$ $\quad \varepsilon_r = 6$

$1 + \chi_e = \varepsilon_r \Rightarrow \chi_e = 5$

Example: Partially-filled parallel plates

A pair of infinite conducting plates at z = 0 and z = 2 m carry equal and opposite surface charge densities of $\pm 2\epsilon_0$C/m^2. Determine $V(2)$ if $V(0) = 0$ and regions $0 < z < 1$ m and $1 < z < 2$ m are occupied by perfect dielectrics with permittivity of ϵ_0 and $2\epsilon_0$, respectively.



Q: Can I use Laplace's equation $\nabla^2 V = 0$ for the inhomogeneous region $0 < z < 2$?

Appr. 1.

$$\begin{cases} \nabla^2 V_1 = 0 & 0<z<1 \quad V_1 = Az+B \\ \nabla^2 V_2 = 0 & 1<z<2 \quad V_2 = Cz+D \\ V_1(0)=0 & V_1(1)=V_2(1) \quad \vec{D}_1 = \vec{D}_2 \end{cases}$$

Appr. 2.

$2\epsilon_0 = \rho_s$

$\vec{D}_1$, $\vec{D}$, $\vec{D}_2$

$-2\epsilon_0 = \rho_s$.

$$\vec{D} = \left(\frac{2\epsilon_0}{2} + \frac{2\epsilon_0}{2}\right)(-\hat{z}) = -2\epsilon_0\hat{z}.$$

$$V(2) - V(0) = -\int_0^2 \vec{E}\cdot d\vec{\ell}.$$

$$= -\left[\int_0^1 -2\,dz + \int_1^2 -1\,dz\right]$$

$$= 3\ [V].$$

$$\vec{E} = \frac{\vec{D}}{\epsilon} = \begin{cases} \frac{-2\epsilon_0}{2\epsilon_0}\hat{z} = -\hat{z} & 1<z<2 \\ \frac{-2\epsilon_0}{\epsilon_0}\hat{z} = -2\hat{z} & 0<z<1 \end{cases}$$

Challenge Question: Spherical shell dielectric

- If we have:

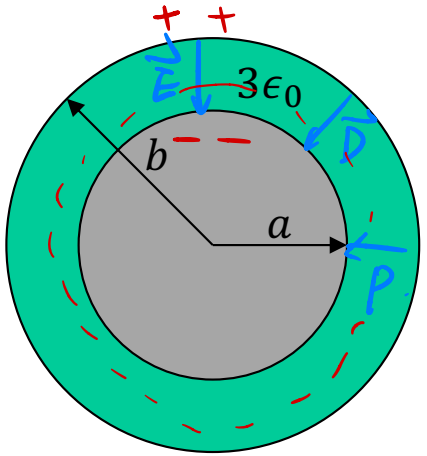
$$\begin{cases} r < a & perfect\ conductor \\ a < r < b & perfect\ dielectric\ \varepsilon = 3\varepsilon_0 \\ r > b & perfect\ conductor \end{cases}$$

with equally and oppositely charged PECs and **D** points radially inward, which is true:

(a) V(b) > V(a)

(b) V(b) = V(a)

(c) V(b) < V(a)



$\vec{E} = -\nabla V$

$\oint\oint \vec{D} \cdot d\vec{S} = Q$

Lecture 9 Summary

- Polarization

$$\vec{\boldsymbol{P}} = \epsilon_0 \chi_e \vec{\boldsymbol{E}}_{tot} = \epsilon_0 \chi_e (\vec{\boldsymbol{E}}_a + \vec{\boldsymbol{E}}_p)$$

  - Homogenous dielectric medium
    - Reduces **E**-field strength by $(1+\chi_e)$
    - $\vec{\boldsymbol{D}}$ has same value as free space
    - $\vec{\boldsymbol{D}}$, $\vec{\boldsymbol{E}}$, and $\vec{\boldsymbol{P}}$ point in the same direction
  - New definition $\vec{\boldsymbol{D}} = \vec{\boldsymbol{P}} + \epsilon_0 \vec{\boldsymbol{E}}$
- Next class
  - Capacitance and Conductance